Tiago Ferrão-Lindqvist

Test & Validation Engineer (Mechanical)

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Professional Summary

Mechanical engineer with 12 years of experience in product design and test and validation of automotive and industrial components. Develop Python tools to validate test data, automate reporting and support engineering decisions, and use SQL to analyse test-result trends. Seeking to move into data engineering, bringing practical experience in data processing, quality checks and engineering problem-solving.

Experience

Test & Validation Engineer | Corvane Automotive Components | Setúbal | Mar 2019–Present

- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms across six servo-hydraulic and electric test rigs; prepare reports sent to OEM customers.
- Developed Python scripts using pandas, NumPy and Matplotlib to read test-rig output files, check each channel against test specifications and generate standard report figures; reduced report preparation from about three days to half a day per test.
- Query the plant's SQL Server test-results database for trend analysis across three product generations.
- Correlate test-rig results with FEA in ANSYS Mechanical and lead 8D root-cause analysis on field failures; proposed a redesigned adjuster rail bracket that cut warranty returns for that part by 40%.
- Maintain MATLAB/Simulink models of the adjuster drive used to select rig load profiles.

Mechanical Design Engineer | Halvik Pumps | Porto | Sep 2014–Feb 2019

- Designed industrial pump casings and impellers in SolidWorks and produced manufacturing drawings with GD&T; owned DFMEA for two product lines.
- Performed structural FEA of casings in ANSYS; supported a change from grey cast iron to ductile iron that reduced material cost by 12% for one casing family.
- Supported conversion of the pump performance test bench to a PLC-controlled setup using Siemens S7.

Education

MSc Mechanical Engineering | Sado Valley Polytechnic | 2014
Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

Skills

Python (pandas, NumPy, Matplotlib); SQL; data validation and reporting automation; MATLAB/Simulink; ANSYS Mechanical; SolidWorks; CATIA V5; LabVIEW; Minitab; design of experiments (DOE); GD&T; DFMEA; 8D problem solving; Excel/VBA; Power BI

Languages

Portuguese (native); English (fluent); Spanish (good)